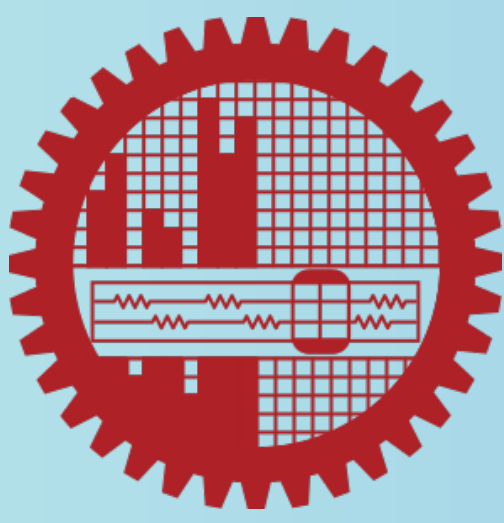




Catastrophic Failures and Explanations

Department of Materials and Metallurgical Engineering
Bangladesh University of Engineering and Technology, Dhaka

Submitted By

2011011 2011057
2011038 2011058
2011040

MORBI BRIDGE TRAGEDY!

Fatigue, Aeroelastic Flutter, Corrosion

October 31, 2022 Death: 141+

Morbi, Gujarat, India

- The 19th-century bridge reopened five days ago, just in time for Diwali and the Gujarati New Year, after undergoing extensive repairs
- The toll operations of the bridge through a contract signed with Oreva on March 7, 2022

Corrosion resulted, in cable from little oiling

1

3

Aeroelastic flutter and allowable capacity assurance



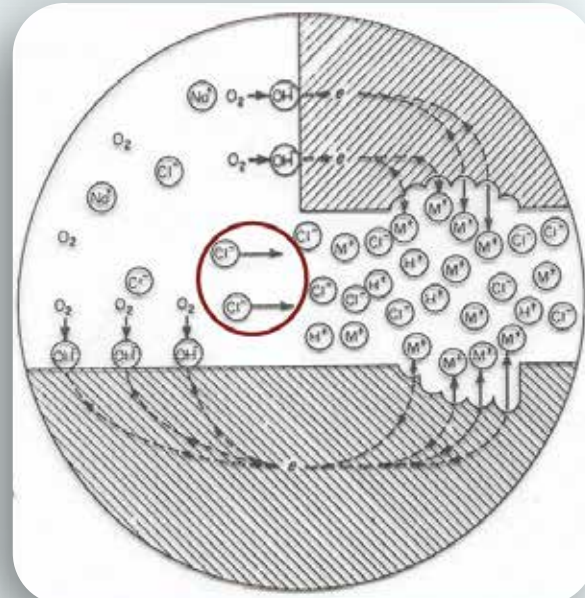
Why it happened?

Fatigue failure in the suspension system

2

4

Four layer Al sheets for flooring



Remedies

Corrosion Protection:	regular inspections, coatings, or corrosion-resistant materials
Wind-resistant Design & Quality :	Aerodynamic shapes or damping systems, can prevent excessive twisting & quality control tests, safe occupancy limits.

Conclusion

Preventative measures require a comprehensive approach, including corrosion protection, proper repairs, quality assurance, wind-resistant design, and adherence to high construction standards.

Skybound Giant's Last Dance, The Hindenburg Disaster

Material selection error, Fatigue

May 6, 1937

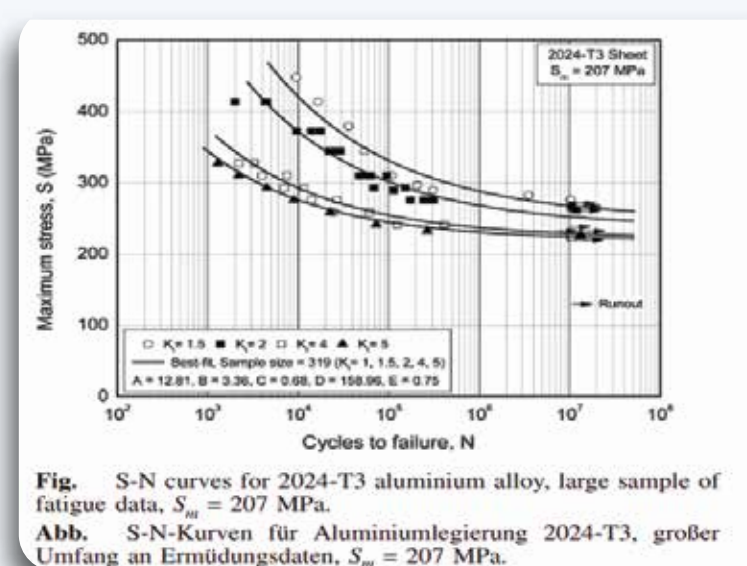
Manchester, New Jersey, USA

Death: 91+

- 803 feet (247 meters) long & 135 feet (41 meters) in dia (Queen of the sky)
- Fire lasted for 34 seconds

Why it happened?

- Used inappropriate aluminum (Al) alloy as outer material
- Employed hydrogen gas (H₂) as the fuel element within the airship
- Possible hypothesis



Remedies

- Fire-resistant materials:** Incorporating non-flammable metals and synthetic fabrics
- Fuel optimization:** Substituting hydrogen with helium for improved safety
- Electrical system refinement:** Implementing spark-resistant technology
- Innovative docking methods:** Implementing better landing system

Conclusion

- Airships: Low emissions of carbon, cost-effective, silent. Enhanced materials for renewed viability and sustainable alternatives in modern transportation.

December 12, 1999

Britanny, France

Death: 42,000 (birds) & 1000 (sea lifes)

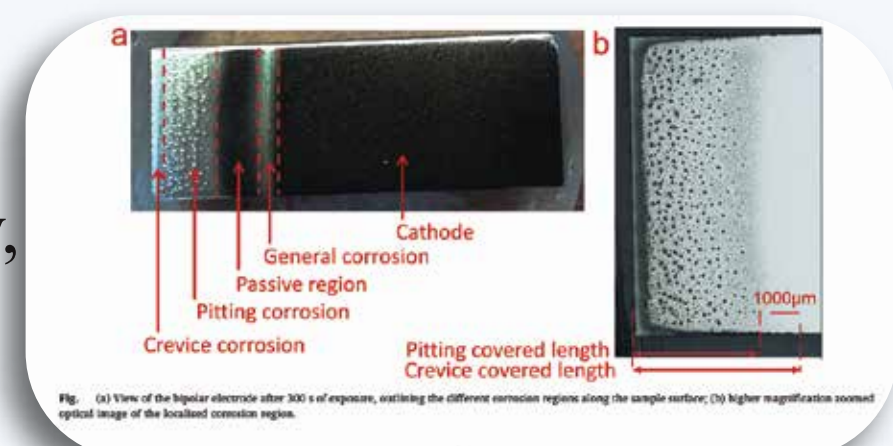
Erika Oil Spill, Europe's Environmental Catastrophe

Corrosion, Bad Design

- A widespread oil sink damaging over 400 km
- Designed as a single-hull oil tanker, launched in 1975 and was owned by a Greek company

Why it happened?

- Structural failure (inaccuracy in assembly, in the quality of welding etc.)
- Tanks were poorly coated
- Weekening of the structure of section no.2 was due to insufficient maintenance and thus rapid corrosion



Remedies

1 Use double-hull tanker instead of single hull

2 Use Duplex SS-grade 2205 (3% Mo)

3 Cathodic protection in the deck

Conclusion

- The Erika tanker disaster underscored the need to protect our oceans and preserve the environment. Simple remedies are crucial for preventing future disasters and safeguarding natural beauty.

